
Topic : Web Graphics Styling Basics

Lab Overview

In this lab, students will learn how to enhance web pages using graphics, images, CSS styling, and semantic HTML elements.

Students will:

- Insert images using HTML
- Create image hyperlinks
- Configure thumbnail images
- Apply CSS background images
- Configure list markers
- Apply visual effects using CSS
- Use semantic HTML5 elements

Required Folder Structure

Create the following structure carefully:

lab5

```
|  
├── index.html  
├── style.css  
└── images  
    ├── beach.jpg  
    ├── flower.jpg  
    ├── background.jpg  
    ├── thumbnail.jpg  
    ├── logo.png  
    └── icon.gif
```

Example Images to Use

File Name	Suggested Image
beach.jpg	Beach / Nature Image
flower.jpg	Flower Image

File Name	Suggested Image
-----------	-----------------

background.jpg	Landscape Background
----------------	----------------------

thumbnail.jpg	Small Flower Image
---------------	--------------------

logo.png	Logo Image
----------	------------

icon.gif	Small Bullet Icon
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PART 1 — Creating Basic HTML Structure

Instruction :

In this activity, students will create the main structure of a basic webpage.

Before starting:

- Create a new folder named lab5
- Inside the folder, create:
 - index.html
 - style.css
 - images folder

Students should understand that HTML is used to create the structure of a webpage.

Observe the webpage carefully after every step.

Objective

Create a simple webpage structure.

Step 1: Create index.html

Create a file named:

index.html

Step 2: Type the HTML Code on your own.

Expected Output

A webpage displaying:

Welcome to My Website

PART 2 — Insert Image Using HTML

Instruction :

In this section, students will learn how to display images on a webpage.

Before typing the code:

1. Save an image named beach.jpg inside the images folder.
2. Make sure the image filename is correct.
3. Question :Why websites use images?
4. Question : What happens if the image path is incorrect?

Students should run the webpage after adding the image.

Objective

Display an image using the `<img>` element.

Step 3: Add Main Image

Insert the following code inside `<body>`:

```

```

Code Explanation

Attribute Description

Src	Specifies image location
Alt	Alternative text for accessibility
Width	Image width
Height	Image height

Expected Output

The webpage should display:

- A beach image
 - Width: 400px
 - Height: 250px
-

PART 3 — Create Image Hyperlink**Instruction :**

Students will now create a clickable image.

- Hyperlinks are not limited to text only.
- Images can also function as navigation buttons.
- Many websites use clickable logos.

Before starting:

1. Save logo.png inside the images folder.
 2. Run the webpage after completing the activity.
 3. Test the image hyperlink by clicking the image.
-

Objective

Create a clickable image.

Step 4: Add Hyperlink Image

```
<a href="https://www.uthm.edu.my">
```

```
  
```

```
</a>
```

When users click the image → Browser opens the UTHM website.

PART 4 — Create Thumbnail Image

Instruction :

In this activity, students will create a thumbnail image.

- A thumbnail is a smaller version of a larger image.
- Thumbnails help reduce webpage loading time.
- Many image galleries use thumbnails.

Before coding:

1. Save flower.jpg and thumbnail.jpg inside the images folder.
 2. Ensure thumbnail.jpg is smaller than flower.jpg.
-

Objective

Create a small image that opens a larger image.

Step 5: Add Thumbnail

```
<a href="images/flower.jpg">  
    
</a>
```

Why Use Thumbnails?

Advantages:

- Faster webpage loading
 - Cleaner layout
 - Saves webpage space
-

PART 5 — Configure Background Image with CSS

Instruction :

Students will now use CSS to apply a background image.

- CSS controls webpage appearance.
- Background images improve webpage design.
- Large images may slow down webpage loading.

Before coding:

1. Save background.jpg inside the images folder.
2. Open the style.css file.
3. Make sure the CSS file is linked correctly inside HTML.

Objective

Apply background image using CSS.

Step 6: Create style.css

Create a file named:

style.css

Step 7: Add Background Image

```
body {  
    background-image: url("images/background.jpg");  
}
```

Step 8: Prevent Background Repetition

```
body {  
    background-image: url("images/background.jpg");  
    background-repeat: no-repeat;  
}
```

Step 9: Resize Background Image

```
body {  
    background-image: url("images/background.jpg");  
    background-repeat: no-repeat;  
    background-size: cover;  
}
```

CSS Explanation

Property	Function
background-image	Sets image as background
background-repeat	Controls image repetition
background-size	Controls image scaling

PART 6 — Configure List Markers**Instruction :**

Students will customize list bullets using image icons.

Questions :

- Why websites use customized bullets.
- The difference between default bullets and image bullets.

Before coding:

1. Save icon.gif inside the images folder.
 2. Refresh the webpage after editing CSS.
-

Objective

Customize list bullets using images.

Step 10: Add HTML List

```
<h2>Services</h2>
```

```
<ul>
```

```
  <li>Website Design</li>
```

```
  <li>Graphic Design</li>
```

```
  <li>SEO Services</li>
```

```
</ul>
```

Step 11: Configure List Marker

```
ul {  
  list-style-image: url("images/icon.gif");  
}
```

Expected Output

The bullet points should display custom icons.

PART 7 — Apply CSS Visual Effects**Instruction :**

In this section, students will enhance webpage appearance using CSS visual effects.

- Modern websites use shadows and hover effects.
- Hover effects improve user interaction.
- CSS effects can make webpages look more professional.

Students should :

1. Hover the mouse over the image.
 2. Observe changes on the webpage.
 3. Modify the opacity value and compare the result.
-

Objective

Enhance webpage appearance using CSS effects.

Step 12: Add Shadow Effect

```
img {  
    box-shadow: 5px 5px 10px gray;  
}
```

Step 13: Add Hover Effect

```
img:hover {  
    opacity: 0.7;  
}
```

Effect Explanation

Effect	Function
box-shadow	Creates shadow around image
opacity	Makes image transparent

PART 8 — Semantic Layout Using Figure**Instruction :**

Students will learn semantic HTML5 elements.

- Semantic elements provide meaning to webpage content.
- `<figure>` groups images and related content.
- `<figcaption>` provides image descriptions.

Questions:

- Why is semantic HTML important?
 - How does semantic HTML improve accessibility?
-

Objective

Use semantic HTML5 elements.

Step 14: Add Figure Section

```
<figure>
  
  <figcaption>
    Beautiful flower image.
  </figcaption>
</figure>
```

Why Use Semantic Elements?

Advantages:

Better webpage structure

Easier for search engines

Improves accessibility

PART 9 — Add Favicon**Instruction :**

Students will configure a favicon.

- A favicon is a small icon displayed on the browser tab.
- Websites use favicons for branding.
- Browsers may cache old favicons.

Before coding:

1. Ensure logo.png exists inside the images folder.
 2. Refresh the browser if favicon does not appear.
-

Objective

Display icon on browser tab.

Step 15: Add Favicon Code

Inside <head>:

```
<link rel="icon"
```

```
href="images/logo.png"
```

```
type="image/png">
```

Expected Output

The browser tab displays a small icon.

REFLECTION QUESTIONS

Question 1

What is the purpose of the alt attribute?

Question 2

Why is PNG suitable for web graphics?

Question 3

What is the function of background-size: cover;?

Question 4

Why are thumbnail images useful in web design?

TROUBLESHOOTING GUIDE

Problem	Possible Cause	Solution
Image not displayed	Wrong file path	Check image location
Background not visible	CSS file not linked	Check <link> tag
Hover effect not working	CSS syntax error	Check brackets and semicolons
Broken image icon	Wrong image filename	Verify image name

Conclusion

In this lab, students successfully learned how to:

- ✓ Insert graphics using HTML
- ✓ Configure image hyperlinks
- ✓ Apply CSS background images
- ✓ Configure list markers
- ✓ Apply CSS visual effects
- ✓ Use semantic HTML5 elements

These skills are essential for designing visually attractive and user-friendly websites.